Putting it all together

DSC 340 Applied Machine Learning

Operationalization [EMC 12]

May 5, 2025

The Plan

- Revisit:
  - Roles in an Analytics Project
  - Data Analytics Lifecycle
  - Developing an Analytics Plan
  - Organizational details
- Reporting and Operationalizing Results
- [if time] Final Project Team/Work Time

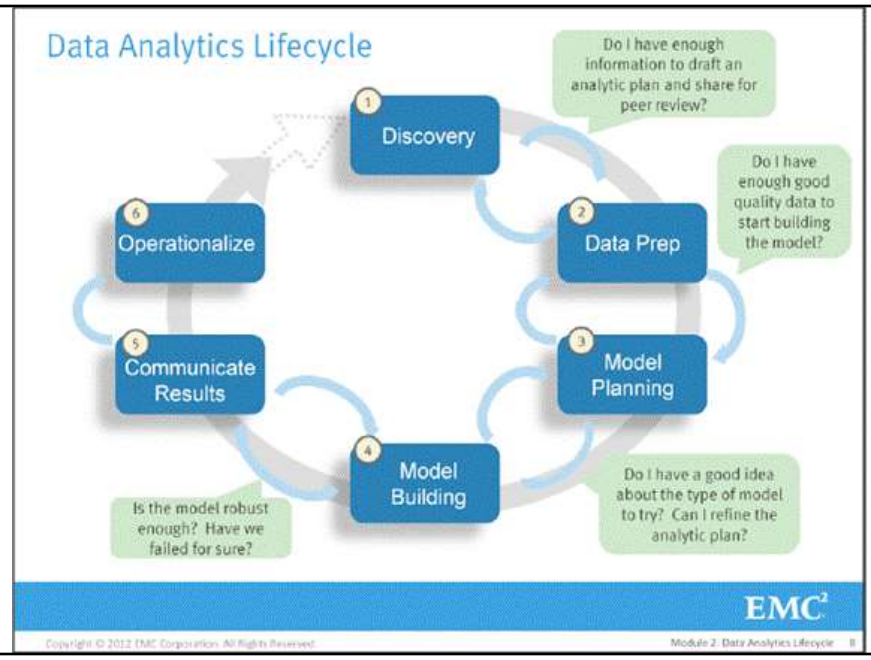
Announcements

- Last Class Days:
  - Wed. May 7 and Fri. May 9: Project Workdays (classroom)
  - Monday, May 12: **Presentations (Lightning Talks)**
  - Tuesday, May 13: Jupyter Notebook (all code + written report) due 11:55 PM

- Comprehensive Final Exam: Friday, May 16th (1:30 - 4:30 PM)
  - Same format as tests
  - About 2x length of tests
    - about 30 MCQs @ 2pts => 60 points
    - about 4 problems/short essays @ 10 pts => 40 points

Data Analytics Roles (Revisited)

- **Organization User:** understands domain area, benefits from work
- **Project Sponsor:** high-level direction from administration
- **Project Manager:** supervises process to meet milestones/objectives
- **BI Analyst:** Business domain expertise, dashboards and reports
- **DB Administrator:** IT support for DB: roles, permissions, installations
- **Data Engineer:** Programming, complex SQL queries, data munging
- **Data Scientist:** The expert in the room for techniques, models, and evaluating result: Choose the right method for the problem at hand.


Data Analytics Lifecycle
Do I have enough information to draft an analytic plan and share for peer review?
1 Discovery
2 Data Prep
Do I have enough good quality data to start building the model?
3 Model Planning
Do I have a good idea about the type of model to try? Can I refine the analytic plan?
4 Model Building
Is the model robust enough? Have we failed for sure?
5 Communicate Results
6 Operationalize
EMC²
Copyright © 2012 EMC Corporation. All Rights Reserved.
Module 2: Data Analytics Lifecycle 8

Developing an analytics plan

- Interview client: include stakeholders
- Do we understand the problem?
- Do we have access to the data we need?
- Do we have the other resources we need?
  - Computational resources
  - Human resources, skillsets
- Develop timetable with landmarks
- Discuss/negotiate/adapt plan with client
- Keep client in the loop (schedule regular meetings/updates)

Organizational details

- Regular collaboration with team
- Shared calendar
- Shared repository with:
  - Raw data
  - Prepared data
  - Code (scripts, notebooks)
  - Visualizations
  - Reports, presentations, etc.
- Data integrity, security, regular backups

Toy example (as always!)

- Problem: Bocce ball turnout
- Predictive model:
  - How many bocce balls?
  - How many individually wrapped snacks?
  - How many bottled beverages of each type?
- **Operationalize**: **dashboard** for event organizers
  - **Prescriptive analytics** vs. **predictive analytics**
  - Place on organization's intranet (accessible locally/VPN)
- Larger project: nationwide rollout if pilot at Centre successful ☺

Summary of Operationalization Phase

- Present final reports
- Assess benefits of analytics project
  - Descriptive: What has happened in the past
  - Predictive: Forecasting for the near future
  - Prescriptive: Support optimal decision making
- Set up pilot program to deploy models
- Larger rollout

Stakeholders

- **Business User:** determines benefits and implications of findings
- **Project Sponsor:** business impact, risks, ROI, how to promote project
- **Project Manager:** project completion, deadline, and budget
- **BI Analyst:** how existing reports and dashboards will be impacted
- **Data Engineer and DBA:** give them analytics code + technical docs
- **Data Scientists:** Share code, explain model to peers, managers, etc.

Deliverables

- **Presentation for Project Sponsors**
  - High-level takeaways for executives
  - Key messages to aid decision-making
  - Focus: clean, easy visuals for presenter to explain and viewer to grasp
- **Presentation for Analytics Team**
  - Describe changes to business processes and reports
  - Visuals include ROC curves, density plots, histograms, etc.
- **Code:** for engineers and others in production environment
- **Technical specifications:** for implementing the code

To Report

- **Project Goals**
- **Main Findings:** Summary, emphasize key messages
- **Approach:** High-level methodology
- **Model Description:** Overview of modeling technique
- **Key Points Supported with Data** (e.g., bar chart)
- **Model Details** (omit for sponsor)
- **Recommendations** (different for sponsor and analysts)